

DDR-Version Portability – Scheduler Parameterization

One channel. **P** = parametric (change a value/reg, plug-and-play). **S** = structural (logic must be added / removed / rewired).

Timing CK values are per speed bin from `timing_reg_file`; DDR5-4800B is our reference.

1. Structural / count parameters

Param	DDR1	DDR2	DDR3	DDR4	DDR5	Kind	Note
N_BANKS	4	4/8	8	16	32	P	count only
N_BG	1	1	1	4	8	S	adds a BG arb level (DDR4+)
BANKS_PER_BG	4	8	8	4	4	P	
BL (burst)	2/4/8	4/8	8	8	16	P	
dqFree = BL/2	1/2/4	2/4	4	4	8	P	DQ heartbeat
N_RANKS	cfg	cfg	cfg	cfg	cfg	P	lane replication
Gear range	1:1	1:1	1:1	1:1..1:2	1:1..1:4	P	phaser slot mask
CA form	parallel	parallel	parallel	parallel	2-cyc 14-bit	S	1-cyc vs 2-beat encoder
ODT	board	on-die	on-die	on-die	on-die	P/S	DDR2+ on-die
AL (posted CAS)	no	yes	yes	yes	no	S	additive latency DDR2-4
Row/Col width	density	density	density	density	density	P	address-map slice

2. Timing constants – which EXIST per generation

Timing	DDR1	DDR2	DDR3	DDR4	DDR5	Kind
tCCD (CAS-CAS)	one	one	one (tCCD)	S/L split	S/L split	S
tRRD (ACT-ACT)	one	one	one (tRRD)	S/L split	S/L split	S
tCCD_L_WR	no	no	no	yes	yes	S
tFAW (4-ACT win)	no	yes (8-bank)	yes	yes	yes	P
tWTR (WR-RD)	one	one	one	S/L split	S/L split	S
tPPD (PRE-PRE)	no	no	no	no	yes	S
tRTRS / tRTW	yes	yes	yes	yes	yes	P
tRC/tRCD/tRP/tRAS/	yes	yes	yes	yes	yes	P
tRFC / tREFI	yes	yes	yes	yes	yes (+tRFCsb)	P/S
ZQ cal	no (OCD)	no (OCD)	yes (ZQCL/CS)	ZQCL/CS	MPC ZQCAL	S

Consequence: DDR3 has NO bank groups $\Rightarrow$ no S/L split $\Rightarrow$ the **BGS/BGD scope tables collapse** and the **BG arb level disappears** (bank $\rightarrow$ rank directly). That is the single biggest structural change going DDR4/5 $\rightarrow$ DDR3.

3. Scheduler logic diff (block by block)

Block	DDR3	DDR4	DDR5	Kind
Bank cells / FSM	8	16	32	P(count)
BGB arb (bank-in-BG)	–	4-way	4-way	S
BG arb (BG-in-rank)	<i>absent</i>	4-way	8-way	S
BGS / BGD scoreboard	collapse	present	present	S
class-split + L3	yes	yes	yes	P
GC + comparators	yes	yes	yes	P(const values)
phaser slot mask	1 slot	≤ 2	≤ 4	P
CA-beat encoder	1-cyc cmds	1-cyc	2-cyc (2 beats)	S
DM / write-CRC path	–	yes	yes	P(present/absent)
on-die ECC assume	–	optional	yes	S

4. Maintenance engine diff

Sub-engine	DDR3	DDR4	DDR5	Kind
REFab	yes	yes	yes	P
REFsb + drain (policy B)	–	–	yes	S(DDR5 add)

Sub-engine	DDR3	DDR4	DDR5		Kind
RFM / RAA (row-hammer)	–	–	yes		S (DDR5 add)
ZQ command form	ZQCL/ZQCS	ZQCL/ZQCS	MPC start/latch	ZQCAL	S
MR_POLL (MR4 TUF)	– (external)	yes	yes		P/S
MR_WRITE (+timing-apply)	basic MRS	MRS	MRW + verify		P/S
INIT sequence	DDR3 seq	DDR4 seq	DDR5 (MPC/train)	seq	S
FREQ_CHANGE (DVFS)	limited	yes	yes (DFI change)	freq-	P
tREFI temp scaling	–	MR4/TUF	MR4/TUF (FCsb)	(+tR-	P

5. Port cost summary (what to touch)

DDR5 → DDR4: params: N_BANKS 32→16, N_BG 8→4, BL 16→8 (dqFree 8→4), gear cap 1:4→1:2, reload timing bin. *Remove:* RFM/RAA engine, REFsb + same-bank drain, tPPD, on-die-ECC assumption. *Rewire:* ZQ (MPC→ZQCL/CS), CA-beat encoder (2-cyc→1-cyc), INIT sequence. BG level **stays**.

DDR4 → DDR3: additionally set N_BG 4→1 ⇒ **delete the BG arb level, collapse BGS/BGD** into bank+rank, and drop all S/L timing splits (single tCCD/ tRRD/tWTR); remove DM/write-CRC; simpler ZQ; simpler init. Gear 1:1 only.

DDR3 → DDR2: N_BANKS 8→4/8, BL 8→4/8; remove ZQ (OCD instead), add AL (posted-CAS / additive latency into the timing model). Still no BG.

DDR2 → DDR1: N_BANKS →4, BL →2/4/8; **remove tFAW** (4-bank, no 4-ACT window), board-level ODT (no on-die), no AL. The arb tree is at its simplest: bank→rank only, no BG, no FAW.

Stays parametric across all three (plug-and-play): bank/rank counts, BL/dqFree, timing CK values (from `timing_reg_file`), gear range, address-map slice, GC+comparator, class-split arb, REFab, FAW window, turnaround (tWTR/tRTW), completion + data-path SRAMs.

The irreducibly structural set (needs RTL, not a knob): presence of the BG arb level + S/L scoreboard, RFM engine, REFsb drain, the 2-cyc CA-beat encoder, and the ZQ/INIT command encodings. Everything else is a parameter or a timing-reg reload.